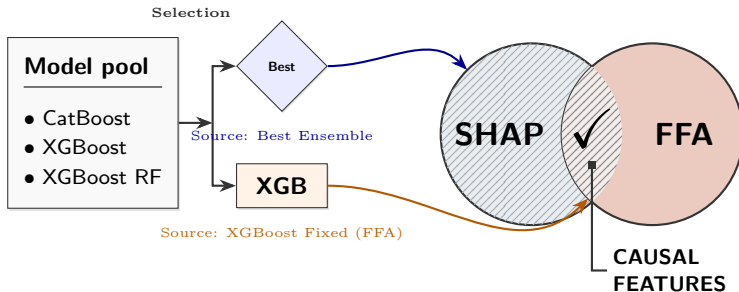


Tournament of Models (ensemble-of-ensembles)

→ $\text{SHAP} \cap \text{FFA} \rightarrow \text{causal consensus}$



Robustness (SHAP): Best-of-three (CatBoost / XGBoost / XGBoost RF) reduces model-specific bias; high SHAP under the selected ensemble strengthens a feature's claim as claims-level signal.

Consistency (FFA): FFA is locked to a standard XGBoost model so feature-frequency baselines stay comparable across temporal holdouts (e.g., 2018 vs. 2019).

Consensus filter (safety gate): In causal terms, the intersection is the gate: high SHAP with negligible FFA can reflect unstable, one-off patterns. Requiring both impact (SHAP) and repeated rule support (FFA) filters out "lucky" correlations in high-dimensional clinical data.